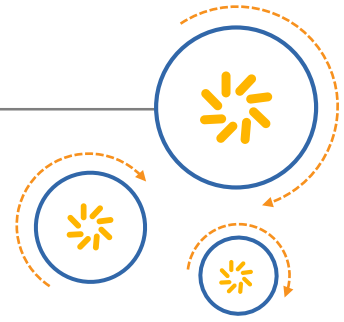




Qualcomm Technologies, Inc.



WCN3620

Device Revision Guide

August 2015

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LM80-P0436-32 Rev. A

Revision history

Revision	Date	Description
A	August 6, 2015	Initial release

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1 Introduction

This device revision guide addresses the WCN3620 WLAN/Bluetooth/FM RF modem solution.

Technical information for WCN3620 devices is primarily covered by the documents listed in [Table 1-1](#). Released WCN3620 documents are posted here:

<https://developer.qualcomm.com/hardware/dragonboard-410c/tools> and are available for download.

Table 1-1 Important WCN3620 documents

Document number	Document title
LM80-P0436-25	WCN3620 Wireless Connectivity IC Design Guidelines
LM80-P0436-26	WCN3620 Layout Guidelines
LM80-P0436-28	WCN3620 Wireless Connectivity Design Example with 2G FEM + External Coupler
LM80-P0436-32	WCN3620 Device Revision Guide (this document)
LM80-P0436-33	WCN3620 Wireless Connectivity IC (Advance Information) Device Specification

1.1 Scope and intended audience

This device revision guide identifies issues with all WCN3620 samples released to date. The following information is included:

- Introduction to this document and its topics (Chapter [1](#))
- Device identification (Chapter [2](#))
 - ❑ Device marking
 - ❑ Hardware revision number
 - ❑ Identification details for each sample type
 - ❑ Sample testing – engineering sample (ES) and customer sample (CS) explanations
 - ❑ Identification of compatible software releases
- Known issues (Chapter [3](#))
 - ❑ Issue description
 - ❑ Impact to system performance
 - ❑ Possible workarounds (what designers should do to minimize the issue’s impact)

This device revision guide is intended for new product developers who are designing, testing, and/or evaluating phones or terminals that include the WCN3620 device.

2 Device Identification

The WCN3620 device can be identified by markings on its top surface and by the contents of an identification register; these identification techniques are described in Section 2.1.

2.1 Device marking

2.1.1 WCN3620 device marking

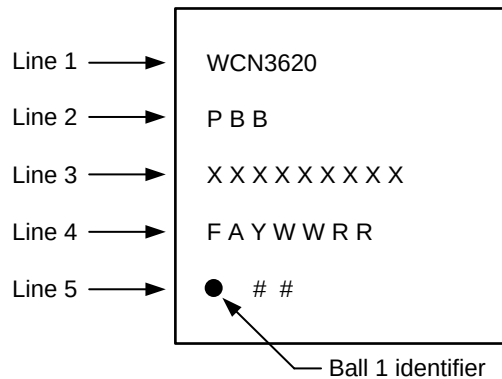


Figure 2-1 WCN3620 part marking (top view – not to scale)

Table 2-1 Part marking line descriptions

Line	Marking	Description
1	WCN3620	Qualcomm product name
2	PBB	P = Product configuration code See Table 2-2 for assigned values BB = Feature code See Table 2-2 for assigned values
3	XXXXXXXXXX	XXXXXXXXXX = traceability information
4	FAYWWRR	F = wafer fab location source code F = B for GF, Fab 7, Singapore A = assembly (drop ball) site code A = C for ASE KH, Taiwan A = D for SCT3, Taiwan Y = single-digit year WW = work week (based on calendar year) RR = product revision – refer to Table 2-2

Line	Marking	Description
5	• XX	• = dot identifying pin 1 XX = traceability information

2.2 Device identification for each sample type

This section provides details for identifying each sample type.

Table 2-2 WCN3620 device identification details

WCN3620 sample type	Product configuration code (P)	Product revision (RR)	Date code (YWW)	Comments
ES1	0	03	–	Engineering sample
ES2	0	04	–	Engineering sample
	0	05	< 315	Engineering sample
CS	0	05	315	Commercial sample with exception ¹
	0	05	≥ 316	Commercial sample
Information for other sample types will be added as they are released.				

1. There is a mix of ES and CS materials with YWW = 315.

- All parts marked with Line3 = 7R30661 and Line5 = 01 and 02 are ES.
- All parts marked with Line3 = 7R30661 and Line5 = 03, 04, 05, and 06 are CS.

2.2.1 Engineering samples

These devices have undergone limited testing and sometimes have significant feature limitations. They are suitable to assist with PCB development, to conduct board-level electrical evaluation tests, and to explore manufacturing considerations. Engineering samples (ES) are not to be used for phone-level qualification.

2.2.2 Commercial samples

These devices have undergone full production-level testing and meet the specifications and features described in the device specification, except as otherwise noted in this document. They have passed device-level qualification. Commercial samples (CS) are suitable to be used for performance testing as well as phone-level production and qualification.

3 Known Issues

3.1 Summary of known issues

All known issues for each revision of the WCN3620 device are summarized in [Table 3-2](#). The text within the *Issue # and description* column provides links to the sections of this document that explain the issues, regardless of the sample type (or types) on which they occur.

NOTE: An *X* in any of the other columns indicates that the issue occurs on the corresponding sample type. A dash (–) in any of the other columns indicates that the issue was fixed on the corresponding sample type.

[Table 3-2](#) also lists the functional area affected by each issue. [Table 3-1](#) describes the functional area abbreviations appearing in [Table 3-2](#).

Table 3-1 Functional area

Functional area	Description
TL	Top-level
WBFM	WLAN, Bluetooth, and FM
W	WLAN
BT	Bluetooth
FM	FM

Table 3-2 Known issues – all sample types and revisions¹

Issue # and description	Functional area (see Table 3-1)	WCN3620 ES1	WCN3620 ES2	WCN3620 CS
		P = 0	P = 0	P = 0
		RR = 03	RR = 05/04	RR = 05
Issue #1 – HTOL digital IO failures	TL	X	–	–
Issue #2 – 11b SEM failure	W	X	–	–
Issue #3 – High amplitude errors (> 0.2 dB)	W	X	–	–
Issue #4 – 19.2 MHz clock phase noise on MSM8x26 platform	W	X	–	–
Issue #5 – WLAN RX max input sensitivity failure at 11b/11n	W	X	X	–
Issue #6 – SCPC output power run-to-run variation	W	X	X	–
Issue #7 – BT EDR Tx continuous in PRBS9 payload cannot be modulated	Bluetooth	X	X	X

1. P and RR values are detailed in [Table 2-2](#).

3.2 Issues – descriptions, impacts, and workarounds

Table 3-3 Issue #1 – HTOL digital IO failures

Description	No longer an issue after the reliability test.
Impact	No impact.
Workaround	No workaround needed.

[Return to Table 3-2](#)**Table 3-4 Issue #2 – 11b SEM failure**

Description	11b SEM failure at target power level.
Impact	The SEM compliant Tx power at 11b is lower than the target specification. WLAN Tx performance is still being optimized with firmware updates.
Workaround	This issue has been fixed in CS via firmware.

[Return to Table 3-2](#)**Table 3-5 Issue #3 – High amplitude errors (> 0.2 dB)**

Description	High amplitude errors (> 0.2 dB).
Impact	Does not meet Wi-Fi RF performance specification.
Workaround	This issue has been fixed in CS via firmware.

[Return to Table 3-2](#)**Table 3-6 Issue #4 – 19.2 MHz clock phase noise on MSM8x26 platform**

Description	The phase noise measured on a MSM8x26 platform is 2 dB – 3 dB worse compared to a MSM8960 platform.
Impact	This degrades the WCN3620 WLAN EVM floor on the MSM8x26 platform by ~ 1 dB.
Workaround	This issue has been fixed in PM8226 CS version.

[Return to Table 3-2](#)**Table 3-7 Issue #5 – WLAN RX max input sensitivity failure at 11b/11n**

Description	WLAN RX max input sensitivity failure at 11b/11n.
Impact	Does not meet WiFi RF performance specification.
Workaround	This issue has been fixed in CS via firmware.

[Return to Table 3-2](#)

Table 3-8 Issue #6 – SCPC output power run-to-run variation

Description	There is a run-to-run variation in SCPC power when measured with power cycle after each run.
Impact	Does not have consistent SCPC power accuracy run-to-run and possibly fail SCPC specification.
Workaround	This issue has been fixed in CS via firmware.

[Return to Table 3-2](#)**Table 3-9 Issue #7 – BT EDR Tx continuous in PRBS9 payload cannot be modulated**

Description	BT EDR Tx continuous in PRBS9 payload cannot be modulated.
Impact	No real user impact since Tx continuous is for FTM and debug purposes
Workaround	Use Tx burst mode when testing PRBS9 payload.

[Return to Table 3-2](#)

A Exhibit 1

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